ARTICLES

Management of Invasive Knotweed

Invasive knotweed has become a monoculture throughout the Upper Delaware River Valley and much of the Northeast.

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Figure 1. Invasive knotweed (Reynoutria spp.) patch along the Upper Delaware River by Joan Jubela, Penn State Master Gardener

Though challenging, through a gardener's patience and a labor of love, it is possible to control this plant and begin to restore native flora. Through a regimen of regular cutting, every year, I consistently battle invasive knotweed along the Delaware River in northeast Pennsylvania. After more than 25 years of trying to control this invasive species, I have acquired a great deal of respect for this herculean plant. I'm resolved not to concoct any fanciful notion of ever eradicating it. Invasive knotweed is here to stay.

Through extensive underground rhizomes, invasive knotweed is a fast grower and spreads quickly. It shades out most native plants and, like many invasive species, becomes an immense monoculture. It colonizes the banks of tributaries large and small, encroaches into wet meadows, and even creeps into woodlands, despite its preference for full sun.

The knotweed I first encountered in the 1990s is no longer the same plant. What was once Japanese knotweed (*Fallopia japonica*) has hybridized, according to Dr. Eric Burkhart, a botanist, ethnobotanist, and agroforester from Shaver's Creek Environmental Center and associate professor at Penn State. *Reynoutria* is a recently assigned taxonomic genus name for two different species of knotweed, *Reynoutria japonica* and *Reynoutria sachalinensis*, known as Japanese knotweed and giant knotweed, respectively. The hybrid of the two is called *Reynoutria* x *bohemica*, or Bohemian knotweed. This hybrid has a slightly different appearance, such as its leaf structure, but for management and control methods, all species can be treated in the same way: cut, cut, cut.

Early in my efforts to control the plant, I learned from landscapers, National Park rangers, and county extension and conservation representatives that, in addition to regular cutting, an application of a glyphosate-based herbicide is used. One common method I have enlisted is to spray the herbicide onto the leaves in late summer. At that time of year, with the knotweed cut down to size, the plant will draw the herbicide into its roots and rhizomatous system. But for a variety of reasons, I elect to use this herbicide sparingly and not every year. For one, special care must be taken when using any herbicide near water.

The chemical agents found in easily purchased products such as Roundup that cause glyphosate to adhere to a plant's leaves are called surfactants. They can be harmful to wildlife that lives in and around water. Many species of both frogs and fish lay eggs near the shoreline, and I don't want to impact the water or cause them harm. Special products for use near water, like Rodeo, are recommended but not readily available. In any case, it is wise to hire contractors trained and licensed in herbicide application from nurseries and landscaping operations, but knowledgeable and well-trained people are not always available.

I cut the plant at least once in late May or early June, then again in early to mid-July. This cycle of cutting usually suppresses flowering. But knotweed easily propagates through fragments too, so I carefully stack this immense biomass into piles above the river's bank for it to thoroughly dry and not be carried downstream by rising water. It is interesting to note that while the flowers of the Bohemian hybrid are sterile, it still grows a vigorous root system.

Initially, when I began to address my knotweed problem, I was not so mindful of any long-range plan or what I intended to accomplish. I had not carefully considered trying to restore what once grew where the knotweed intruded. I was only fueled by the desire to be rid of its towering, dense mass of bamboo-like canes, easily reaching 15 feet high. In cutting, large swaths were created, and other plants vied to fill the void. An essential question arose: What will revegetate the site; what successional pathway will take the place of the knotweed? It's not an easy problem to solve. I've spent a lot of time weed-whacking Japanese stilt grass because I did not consider this question years ago. Today, I am much more conscious and thoughtful.

Dr. Burkhart recommends re-planting areas with vigorous native woody shrubs and trees such as dogwoods and willows. Live staking is reported by many sources to be a good option. Knotweed is less vigorous in shade, so establishing an overstory or other competition will help to suppress growth.

In addition to planting new species, recognizing and encouraging plants that are growing in the vicinity is a viable approach in an overall strategy of restoration. One species in my case that proved indomitable, despite deer browse, is common pokeweed (*Phytolacca americana*). Reaching as high as 10 feet, it has played a significant role in overshadowing the knotweed.

Figure 2. St. John's wort (*Hypericum ascyron*) along the river by Joan Jubela, Penn State Master Gardener

Sometimes the notion of "encouraging" can simply mean not mowing down a plant with a lawnmower, as was the case with giant St. John's wort (*Hypericum ascyron*). For years, a single tall plant growing atop the river's bank in a patch of tall fescue intrigued me. The large green buds never bloomed, but when any mowing occurred nearby, I made sure it got a wide berth. Finally, one year, it bloomed, producing a stellar, yellow flower with attractive, bewhiskered-like stamens. Over the course of a few years, several giant St. John's wort have grown down the bank almost to the shoreline of the river. It produces new plants every season. I continue to make sure each one has lots of space.

Figure 3. Blue vervain (*Verbena hastata*) with bumble bee by Joan Jubela, Penn State Master Gardener

Also, in subsequent seasons, with the knotweed kept at a height of no more than 3 or 4 feet, many interesting natives emerge. These include sneezeweed (*Helenium autumnale*), Alleghany monkey flower (*Mimulus ringens*), wild bergamot (*Monarda fistulosa*), and blue vervain (*Verbena hastata*). In a diverse ecology, lots of plants grow in a floodplain. While many aren't native, I happily tolerate some of them—common field mustard (*Brassica rapa*) is one. I've attempted my own plantings, too, like partridge pea (*Chamaecrista fasciculata*) and Joe Pye weed (*Eutrochium fistulosium*), both grown from seed. Initially, the Joe Pye germinated and grew in pots on my deck. It is Pennsylvania's tallest wildflower, reaching 9 feet. In its second year, it bloomed profusely, and in August, it reached nearly five feet. Until it reaches full height, I am protecting this drift with multi-purpose netting.

I've begun to incorporate grasses to replace knotweed, too. Prairie dropseed (*Sporobolus heterolepis*) grows nearby, but when I planted this grass, it proved to be a slow grower. Wild rye (*Elymus virginicus*) and switchgrass (*Panicum virgatum*) are two native grasses I plan to try; at least, I'm researching them. Many invasives continue to crop up as well. Mugwort (*Artemisia vulgaris*) is difficult to control, but it can be smothered. More recently, small carpet grass (*Arthraxon hispidus*) is proving to be very aggressive, and I'm observing it closely. Overall, many species are proliferating, and the invasive knotweed is present but no longer a monoculture. What's exciting for me is that every season is different.

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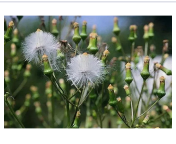
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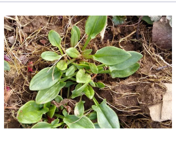
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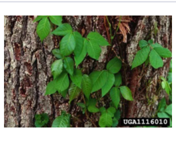
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